

want to skip certain cameras.

Edit key (TV/AV). To change the camera hold-up time for 'auto mode,' enter 'edit' mode first by pressing this key once and then set the desired time by pressing the time up/down key as described above. To exit edit mode after setting the time, press the same key on the remote once again. The controller goes back to manual mode.

Testing and troubleshooting

After assembling the circuit without microcontroller, apply the power using 'on'/'off' switch S1. Check the DC voltage at pin 3 of regulator IC7. It should be +5V. Check the same voltage at pins 1 and 20 of the base of IC1.

Now, switch off the power, insert the microcontroller in its base and recheck all the connections from the PCB to the LCD and IR sensor TSOP1738. On applying the power, page-1 and page-2 text should display on the LCD screen. Using the remote, switch on the switcher and change the functional status on the LCD by referring to CCTV switcher remote commander (shown in Fig. 7) and LCD screenshots (shown in Fig. 8).

Adjust VR1 for maximum contrast on the LCD. If you don't get any output on the LCD, troubleshoot as follows:

1. No output on the LCD. Switch off the power and check continuity between the LCD and various pins of IC1 and regulator IC as shown in the schematic diagram. If the connections are OK, switch on the power, vary VR1 and check the voltage at pin 3 of the LCD (0V to 5V).

2. No backlight on the LCD. Check the voltage at pin 15 (+5V) and pin 16 (0V) of the LCD.

3. Page 1/page 2 is shown on the screen but the remote commander doesn't work (nothing happens on pressing functional keys of the remote). Check the supply at pin 2 of the IR sensor. It should be +5V. If +5V is available at pin 2, check the voltage at pin 3 of the IR sensor. If there is no voltage

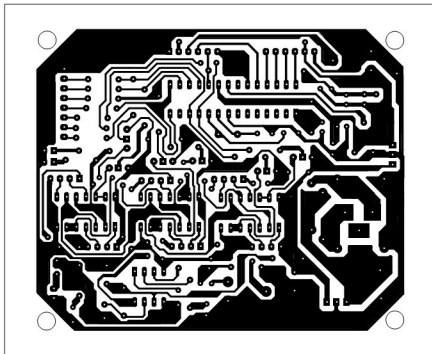


Fig. 8: Actual-size, single-side PCB for the remote-controlled 6-camera CCTV switcher

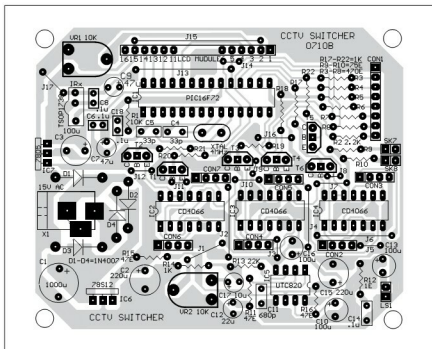


Fig. 9: Component layout for the PCB

at pin 3, replace the IR sensor or check the PCB, etc for shorting.

Once the LCD shows page-1 and page-2 text properly and the remote commander is working, switch off the power and then switch it on again

using S1. Now the LCD should show page-1 and page-2 as shown in Fig. 3. At the same time, pin 14 of the MCU should go high (+5V) and the power standby LED should glow. Now press 'Power' key on the remote. Pin 14